



DAV — Voice-Assisted Design

User Manual

DAV is an inclusive development designed to promote the participation of people with motor disabilities in 2D/3D design.

The goal of the project is to enable people with motor disabilities to create and modify 3D models, drawings, and parts through spoken instructions. In doing so, it seeks to offer a voice-based alternative that complements the keyboard and mouse within the CAD environment, broadening opportunities for participation and promoting technological accessibility.

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Introduction

Introduction

The DAV (Voice-Assisted Design) project is a tool designed to facilitate interaction with CAD systems through voice commands, providing a more accessible and integrated design experience.

Getting started

The system is based on FreeCAD 1.1 (tested with version 1.1.3). Make sure it is installed to ensure compatibility with our voice control layer. Installation guides are in *Dav/docs/*.

Interface and commands

Graphical interface: developed with PySide, it shows the object tree, the buttons of each context and lets you configure preferences.

Voice control: uses the Vosk engine. The system recognizes specific commands that are translated into actions within FreeCAD.

Display modes: supports light and dark modes, integrated with the software's native settings.

Main features

- 2D and 3D drawing (Draft, Part, PartDesign, Sketcher, Assembly and TechDraw workbenches).
- Voice navigation between files, projects and printing.
- Voice-based object selection with visual feedback in the object tree.
- Voice corrections: undo, redo and delete objects with confirmation.
- Step-by-step guided examples inside DAV.
- Multilingual support (Spanish, English, Portuguese).

Assistance and preferences

Use the preferences panel to adjust the speech model based on your system's available RAM (small models for low consumption, large for higher accuracy).

General operation of voice navigation

The DAV system operates through hierarchical contexts (level navigation):

1. **Initial access:** from the Base level, say «explorer» or «file» to enter the Explorer submenu, «selection» for object selection, «workbench» for the workbenches or «standard view» for the views.
2. **Submenus (enter):** when you say a category (e.g. «file», «edit», «additive»), the system descends to that level and enables only its commands.
3. **Executable commands:** perform the action in FreeCAD when spoken.
4. **Go up a level:** in any subcontext, saying «up», «back» or «go back» returns to the upper level.
5. **View status:** saying «context» or «where am I» lists the available options on screen.

How numbers are dictated

- Numbers from 0 to 99 are said naturally («thirty two»). From 100 on they are said digit by digit («two five zero» = 250).
- Decimals use «point»: «one point five». Negatives use «minus».
- Each value is said and confirmed with «okay» (or «send», «accept») in a separate phrase.

Global and navigation commands

They can be said in any context, without navigating to their submenu.

Navigation and confirmation

Icon	Voice commands (English)	What does it do?	Requirements
Up up , go up , go back , return up		Goes up one level in the context tree.	None.
Context context , where am i , what can i say , available options show_context		Lists the submenus and commands available at the current level.	None.
Confirm send , enter , accept , confirm send		Confirms the value or option dictated in a voice dialog.	An open voice dialog.
Cancel cancel , discard , never mind , abort cancel		Discards the phrase in progress or closes the voice dialog.	None.

Correction, measure, view and panel

Icon	Voice commands (English)	What does it do?	Requirements
 Undo undo , undo change undo		Undoes the last change in the document.	Action history available.
 Redo redo , redo change redo		Redoes the last undone change.	Action history available.
 Delete last delete last , remove last , delete last object deleteLast		Deletes the last object created, with voice confirmation.	Active document.
 Delete object delete object , remove object deleteObject		Deletes an object chosen by voice, with confirmation.	Active document.
 Delete broken delete broken , remove broken , clean broken , clean errors deleteBroken		Removes broken or errored objects, with confirmation.	Active document.
 Measure / dimension measure , measure distance , dimensioning , tape measure CreateDimension		Creates a dimension between the selected elements.	Selected object.
Help help , info , options ayuda		Shows help for the current context.	None.
Move view move , move view , move camera , pan MoveView		Centers the camera on the given point and pans it.	Active 3D view.
 Preferences preferences , settings preferences		Opens the DAV preferences window (voice model, language).	None.
Minimize panel minimize , hide panel hide_dav_panel		Minimizes the DAV panel.	None.
Maximize panel maximize , show panel show_dav_panel		Shows the DAV panel again.	None.

The **standard views and zoom** (front, top, iso, zoom in, zoom out, fit all...) can also be said from any context; they are in the *Standard Views* section.

1. Explorer Dictionary

Files, editing, printing, windows and general tools. Enter by saying «explorer».

Icon	Voice commands (English)	What does it do?	Requirements
	refresh reload , update refresh	Recomputes and refreshes the tree and the 3D view.	Active document.
	screenshot photo , take photo , capture screenshot	Opens the dialog to save an image of the 3D view.	Active 3D view.
	text document textdoc , text file , document textdoc	Creates a note or text document inside the tree.	Active document.
	unlink remove link , detach link unlink	Breaks the object's link to its original source.	Selected link.
	freeze lock , immobilize freeze	Freezes the object to prevent automatic recomputation.	Selected object.
	all instances select all instances , allinstances allinstances	Selects every shared instance of the object.	Object with several instances in the scene.
	variable set variables , var set , variableset variableset	Creates a VarSet to store global variables and expressions.	Active document.
	file Group — enter by saying: file , files , folder	Document management with FreeCAD's native dialogs.	None.
	new new file , create , create file new	Creates a new blank document.	None.
	open open file open	Opens the system file browser to load a project.	None.
	save save file save	Saves the active document to its current path.	Active document already saved.
	save as save document as saveas	Saves the active document under a new name or path.	None.
	save copy duplicate file savecopy	Saves a copy without changing the working file.	None.
	revert restore revert	Discards changes and restores the last saved version.	Active document already saved.
	merge merge projects merge	Merges the contents of another FCStd file into the document.	None.
	import import file import	Imports external 2D/3D models (STEP, IGES, DXF, OBJ...).	None.
	export export file export	Exports the selection to external 2D/3D formats.	Selected object.

Icon	Voice commands (English)	What does it do?	Requirements
	recent recent files recent	Shows the most recently opened documents.	None.
	load image open image loadimage	Inserts an image as a reference plane in 3D space.	None.
	project Group — enter by saying: project , projects	Open, save and export with voice windows (no native dialogs).	None.
	new new project , new file , create new	Creates a new project (same as file «new»).	None.
	open open project , open file , find file open	Browses folders by voice (next, previous, open, up) and opens the chosen file.	None.
	save save project , save file save	Saves the document; if it is new it asks for folder and name by voice.	None.
	export export project , export file export	Chooses format (STEP, IGES, STL, OBJ, DXF), folder and name, and exports the selection or everything visible.	None.
	3d print 3d printing , three d print , export for printing print3d	Chooses format (3MF, STL, OBJ), folder and name, and exports the solid parts ready for 3D printing.	Selected part or solid.
	edit Group — enter by saying: edit , editing , modify	Undo, clipboard, placement and properties.	Active document.
	undo revert , go back undo	Reverts the last action performed.	Action history available.
	redo step forward redo	Reapplies the last undone action.	Action history available.
	cut cut	Copies to the clipboard and removes the selected object.	Selected object.
	copy copy	Copies the current selection to the clipboard.	Selected object.
	paste paste	Pastes the clipboard contents into the active document.	Something copied to the clipboard.
	duplicate clone duplicate	Creates an identical, independent copy of the selection.	Selected object.
	select all grab all selectall	Selects every object in the document.	Document with objects.
	delete remove , erase delete	Deletes the selected objects from the document.	Selected object.
	placement position , set position placement	Opens the numeric position and rotation dialog.	Selected 3D object.
	transform move transform	Enables the 3D manipulators to move and rotate the object.	Selected 3D object.

Icon	Voice commands (English)	What does it do?	Requirements
	align alignment align	Opens the tool to align two or more objects.	Two or more selected objects.
	preferences settings preferences	Opens FreeCAD's general preferences panel.	None.
	properties details properties	Focuses the object's properties panel (Data/View).	Selected object.
	send to python python console sendtopython	Sends the object reference to the Python console.	Selected object.
	edit mode modify mode editmode	Enters the object's interactive edit mode.	Editable object (sketch, mesh, etc.).
	print Group — enter by saying: print , printing , printer	Physical printing and PDF export of the view or sheet.	Active document.
	print printing , printer , print document print	Opens the print dialog for the current view or sheet.	Active 2D/3D view or TechDraw sheet.
	pdf pdf	Exports the current view or sheet to a vector PDF.	Active 2D/3D view or TechDraw sheet.
	windows Group — enter by saying: windows , window	Closes tabs, documents or the application.	None.
	close close window close	Closes the active tab or document.	Active document.
	close all close all windows closeall	Closes all open project tabs.	Active document.
	quit exit , exit program quit	Closes FreeCAD completely, asking to save first.	None.
	expressions Group — enter by saying: expressions , expression	Copies and pastes expression and formula references.	Active document.
	copy active document copy document copyactdoc	Copies the active document reference for use in formulas.	Active document already saved.
	copy all documents copy all copyalldoc	Copies every expression reference in the tree.	Active document.
	copy selected copy selection copyselected	Copies the formula path of the selected parameter.	Selected object.
	paste expression paste expressions pasteexpr	Pastes and binds the copied expression into the selected property.	Property open for editing.
	tools Group — enter by saying: tools , tool , utilities	Measure, customize and advanced utilities.	Active document.
	measure measure distance , dimensioning , tape measure CreateDimension	Activates the interactive ruler to measure distances and angles.	Active 3D view.

Icon	Voice commands (English)	What does it do?	Requirements
	clarify selection clear selection clarifyselection	Shows a menu that sorts the selection by geometry type.	Selected object.
	demo mode demonstration demomode	Rotates the camera continuously to present the model.	Active 3D view.
	customize settings customize	Opens the dialog to customize shortcuts, toolbars and commands.	None.
	edit parameters parameters editparameters	Opens FreeCAD's advanced parameter editor.	None.
	project utility project tools projectutil	Opens the utility that unpacks and repairs the structure of an .FCStd file.	None.
	structure Group — enter by saying: structure , structure toolbar , structure bar	Structural containers and links.	Active document.
	part component , piece , element part	Creates a Part container to group bodies or solids.	Active document.
	group new group , collection , folder newgroup	Creates a generic group folder in the tree.	Active document.
	link Group — enter by saying: link , connect , associate	Creating and managing linked instances.	Active document.
	make link create link , generate link , new link makelink	Creates a lightweight link (App::Link) to an existing object.	Selected object.
	relative link linked relative , attach relative relativelink	Creates a link to a relative sub-element.	Selected object.
	import link bring link , load link importlink	Imports an external object as a link.	Active document.
	import all load all links , bring all links importalllinks	Imports every linked element into the active file.	External links in the project.
	replace link substitute link , change link replacelink	Replaces a link with its original object.	Selected object.
	link groups group links , link set , link cluster linkgroups	Groups several links in a container.	Active document.
	examples Group — enter by saying: examples , example , learn	User manual and step-by-step guided examples.	None.
	manual reference , guide , user manual manual	Opens the user manual (PDF) in the DAV language.	None.
	examples example , demos , tutorial demos	Opens the guided example picker (sketch, draft, die, washer, bolt-nut, scissors...).	None.

2. Standard Views (StdView)

Views, zoom, camera, appearance, panels and tree. Enter by saying «standard view»; views and zoom can also be said from any context.

Icon	Voice commands (English)	What does it do?	Requirements
	appearance Group — enter by saying: appearance , visual appearance , look	Colors, textures and appearance of objects.	Active 3D view.
	appearance look , aspect appearance	Opens the appearance panel of the selected object.	Selected object.
	facecolor background , Surface color , Part color facecolor	Assigns a different color to each face of the part.	Selected object.
	random color randomized color , random shade randomcolor	Assigns a random color to the object.	Selected object.
	texture mapping texture projection , texture placement texturemapping	Applies or edits the object's texture mapping.	Selected object.
	camera Group — enter by saying: camera , view camera	Camera projection type.	Active 3D view.
	orthographic ortho orthographic	Switches to orthographic view (no perspective).	Active 3D view.
	perspective perspectiveview perspective	Switches to perspective view.	Active 3D view.
	clipping Group — enter by saying: clipping , clip , clipping plane	Clipping plane of the 3D view.	Active 3D view.
	clipping clip , clipping plane clipping	Enables the clipping plane to see inside parts.	Active 3D view.
	draw styles Group — enter by saying: draw styles , drawing styles , display styles	Drawing style of the geometry.	Active 3D view.
	asis as is , original styleasis	Shows the model as defined.	Active 3D view.
	points point mode , dots points	Shows points only.	Active 3D view.
	wireframe wire , wire frame wireframe	Shows edges only (wireframe).	Active 3D view.
	hidden hidden line , hidden lines hiddenline	Shows edges hiding those behind surfaces.	Active 3D view.
	noshading no shading , without shading noshading	Shows faces without shading.	Active 3D view.
	shaded shaded view , solid shaded	Shows the shaded solid.	Active 3D view.

Icon	Voice commands (English)	What does it do?	Requirements
	flat flat lines , flatlines flatlines	Shows flat faces with their edges.	Active 3D view.
	material Group — enter by saying: material , materials	Library materials.	Active 3D view.
	material set material , apply material , change material material	Assigns a library material to the object.	Selected object.
	overlay Group — enter by saying: overlay , overlays , overlay view	Panels overlaid on the 3D view.	Active 3D view.
	bottom overlay overlay bottom bottom	Overlays the bottom panel.	Active 3D view.
	float floating overlay , float overlay float	Makes the panel floating.	Active 3D view.
	left overlay overlay left left	Overlays the left panel.	Active 3D view.
	right overlay overlay right right	Overlays the right panel.	Active 3D view.
	axis axis cross , show axis axis	Shows or hides the axis cross.	Active 3D view.
	navigation toggle navigation , navigation panel navigation	Toggles the navigation panel.	Active 3D view.
	toggle toggle overlay , show overlay toggle	Toggles the panel overlay.	Active 3D view.
	panels Group — enter by saying: panels , panel , view panels	Open, dock and close FreeCAD panels.	Active 3D view.
	panel panel view panel	Shows or hides the main panel (DAV / combo view).	Active 3D view.
	dock dock view dock	Docks the current view into the main window.	Active 3D view.
	fullscreen toggle fullscreen fullscreen	Toggles full screen.	Active 3D view.
	undock undock view undock	Undocks the current view into its own window.	Active 3D view.
	dag view dagview	Opens the DAG view (dependency graph).	Active 3D view.
	combo view comboview	Shows or hides the combo panel (tree and tasks).	Active 3D view.
	selection view selectionview	Opens the selection view panel.	Active 3D view.
	task view tasks tasks	Opens the tasks panel.	Active 3D view.

Icon	Voice commands (English)	What does it do?	Requirements
	property view properties properties	Opens the properties panel.	Active 3D view.
	python console console console	Activates the Python console.	Active 3D view.
	report view report report	Opens the report message view.	Active 3D view.
	tree view treeview	Shows or hides the model tree.	Active 3D view.
	status bar statusbar	Shows or hides the status bar.	Active 3D view.
	saved views Group — enter by saying: saved views , saved view , bookmarked views	Saved camera views.	Active 3D view.
	clear erase , remove , reset clear	Clears the frozen views.	Active 3D view.
	freeze lock , hold , fix freeze	Freezes the current camera view.	Active 3D view.
	restore recover , bring back restore	Restores the chosen frozen view.	Active 3D view.
	recall remember , retrieve , call back recall	Recalls the stored working view.	Active 3D view.
	load open , fetch , bring load	Loads frozen views from a file.	Active 3D view.
	save keep , record save	Saves the frozen views to a file.	Active 3D view.
	store archive , register , keep view store	Stores the current view as the working view.	Active 3D view.
	standard views Group — enter by saying: standard views , standard view , basic views	Standard views, zoom and framing. They can be said from any context.	Active 3D view.
	bottom below , down , lower bottom	Bottom view.	Active 3D view.
	box zoom window zoom , zoom window boxzoom	Zooms to a chosen rectangle.	Active 3D view.
	new view create view newview	Opens a new 3D view window.	Active 3D view.
	dimetric dimetric view , dimetric projection , dimetric perspective dimetric	Dimetric view.	Active 3D view.
	fit all fit , fitview , zoomfit fitall	Fits the whole model in the view.	Active 3D view.

Icon	Voice commands (English)	What does it do?	Requirements
	fit selection fit selected , zoom selection fitselection	Fits the selection in the view.	Selected object.
	front forward front	Front view.	Active 3D view.
	full screen full screenmode fullscreen	Toggles the view to full screen.	Active 3D view.
	home default view , reset view home	Returns to the home view.	Active 3D view.
	isometric iso isometric	Isometric view.	Active 3D view.
	left left view , left side , from the left left	Left view.	Active 3D view.
	rear back , behind rear	Rear view.	Active 3D view.
	right right view , right side , from the right right	Right view.	Active 3D view.
	top above , upper top	Top view.	Active 3D view.
	trimetric trimetric view , trimetric projection , trimetric perspective trimetric	Trimetric view.	Active 3D view.
	zoom in zoom in view zoomin	Zooms in.	Active 3D view.
	zoom out zoom out view zoomout	Zooms out.	Active 3D view.
	stereo Group — enter by saying: stereo , stereoscopic , stereo view	Stereoscopic viewing modes.	Active 3D view.
	camera position view camera position , issue cam position camerapos	Prints the camera position to the console.	Active 3D view.
	stereo columns interleaved columns , column stereo mode stereocolumns	Stereo with interlaced columns.	Active 3D view.
	stereo rows interleaved rows , row stereo mode stereorows	Stereo with interlaced rows.	Active 3D view.
	stereo off disable stereo , turn off stereo stereooff	Turns stereo mode off.	Active 3D view.
	stereo quad quad buffer , quad buffer stereo stereoquad	Quad-buffer stereo.	Active 3D view.
	stereo anaglyph red green stereo , anaglyph mode stereoanaglyph	Red-green anaglyph stereo.	Active 3D view.

Icon	Voice commands (English)	What does it do?	Requirements
	toolbars Group — enter by saying: toolbars , toolbar , view toolbars	Show or hide toolbars.	Active 3D view.
	clipboard copy , paste , cut clipboard	Toggles the clipboard toolbar.	Active 3D view.
	edit editing edit	Toggles the edit toolbar.	Active 3D view.
	file file	Toggles the file toolbar.	Active 3D view.
	help toolbar toolbar help toolbarhelp	Toggles the help toolbar.	Active 3D view.
	views individual views , view toolbar views	Toggles the views toolbar.	Active 3D view.
	lock lock toolbars , unlock toolbars lock	Locks or unlocks the toolbars.	Active 3D view.
	macro macros macro	Toggles the macro toolbar.	Active 3D view.
	structure structure	Toggles the structure toolbar.	Active 3D view.
	view view	Toggles the view toolbar.	Active 3D view.
	workbench workbench	Toggles the workbench toolbar.	Active 3D view.
	tree Group — enter by saying: tree , model tree , document tree	Behavior of the model tree.	Active 3D view.
	collapse collapse tree , fold collapse	Collapses every item of the active document in the tree.	Active 3D view.
	preselection preselect , preview selection preselection	Preselects in the 3D view the object hovered in the tree.	Active 3D view.
	record selection save selection recordselection	Records the tree selection so you can go back and forward.	Active 3D view.
	single expand expand one , open one singleexpand	Shows only the active document in the tree and expands its items.	Active 3D view.
	syncplacement sync placement , align placement syncplacement	Adjusts placement automatically when dragging objects across coordinate systems.	Active 3D view.
	sync selection match selection syncselection	Expands in the tree the object selected in the 3D view.	Active 3D view.
	syncview sync view , match view syncview	Switches to the 3D view that contains the selected object.	Active 3D view.
	visibility Group — enter by saying: visibility , visible , show hide	Show, hide and select by visibility.	Active 3D view.

Icon	Voice commands (English)	What does it do?	Requirements
	hide objects hide objects , hide all , conceal objects hideobjects	Hides all objects.	Document with objects.
	hide hide selection , hide selected , conceal hide	Hides the selected objects.	Selected object.
	all links select all links , every link alllinks	Selects every link to the selected object.	Selected object.
	linked linked object , linked target , go to link linked	Selects the linked object.	Selected link.
	linked final final link , deepest link linkedfinal	Selects the final object of a link chain.	Selected link.
	selection back back selection , previous selection , go back selection selback	Goes back to the previous selection.	Active 3D view.
	bounding box boundary box , bbox , show bounding box boundingbox	Shows the bounding box of the selection.	Selected object.
	selection forward forward selection , next selection , go forward selection selforward	Goes forward to the next selection.	Active 3D view.
	select visible select visible objects , select visible only selectvisible	Selects the visible objects.	Document with objects.
	show objects show all , reveal objects showobjects	Shows all objects.	Document with objects.
	show reveal , show selection , show selected show	Shows the selected objects.	Selected object.
	toggle all toggle everything toggleall	Inverts the visibility of every object.	Document with objects.
	select ability toggle selectability , selectable selectability	Toggles whether the selected objects can be picked with the mouse.	Selected object.
	transparency toggle transparency , transparent transparency	Toggles the selection's transparency.	Selected object.
	toggle toggle visibility , switch visibility toggle	Toggles the selection's visibility.	Selected object.
	align to selection align camera to selection aligntoselection	Aligns the view with the selection.	Selected object.

3. Line Attributes (LineAttributes)

Line style, width and color for TechDraw views.

Icon	Voice commands (English)	What does it do?	Requirements
	attributes Group — enter by saying: attributes , properties , features	Pick and apply line attributes in two steps.	Active TechDraw sheet.
	select line attributes select attributes , choose attributes select	Step 1: chooses the attributes (style, width, color) to apply.	Active TechDraw sheet.
	change line attributes change attributes , edit line attributes , modify attributes change	Step 2: applies the chosen attributes to the selected lines.	Lines selected in the view.

4. Selection Tool (Selection)

Steps through document objects by voice, without a mouse: the object reached is highlighted in the 3D view and in the tree. Enter by saying «selection».

Icon	Voice commands (English)	What does it do?	Requirements
	advance skip , next , next object next	Moves to the next object in the list and highlights it.	Document with objects.
	go back previous , previous object , previous item previous	Moves back to the previous object in the list and highlights it.	Document with objects.
	all everything , select all , select all objects selectall	Selects every object in the document.	Document with objects.
	none nothing , uncheck , uncheck all deselectall	Clears the selection, leaving nothing selected.	None.
	which this , current , current object current	Says the name of the current object.	Document with objects.
	how many count , how many objects count	Reports how many objects the document has.	Active document.
	remove delete , delete object , delete element delete	Deletes the selected object (first next/previous, then delete).	Selected object.
	search by spelling search spelling , spell search , search object spell	Spell an object's name and selects the closest match.	Document with objects.
	paint object paint part , color object , change color color	Paints the selected objects with a color picked by voice.	Selected object.
	object material part material , pick material , choose material material	Gives the selection a library material picked by voice.	Selected object.

5. Workbench: Assembly

Enter with «workbench» → «assembly».

Icon	Voice commands (English)	What does it do?	Requirements
	new assembly create assembly create	Creates a new assembly.	Active document.
	new part insert part newpart	Creates an empty part (Part with its Body) inside the assembly, with no dialogs.	Active assembly.
	insert link link part link	Inserts an existing document component as a link.	Active assembly.
	solve solve assembly solve	Solves the assembly and applies every joint.	Active assembly.
	exploded view view	Creates the radial exploded view: you dictate the spacing in mm.	Active assembly.
	simulation create simulation simulation	Creates a motion simulation of the assembly.	Assembly with two parts or chosen faces/edges.
	bill of materials bom bom	Generates the bill of materials (index, name, description, file, quantity).	Active assembly.
	preferences settings preferences	Opens the Assembly workbench preferences.	None.
	ground toggle grounded , ground part , anchor part grounded	Grounds a part chosen by voice so the solver does not move it.	Active assembly.
	fixed joint lock parts fixed_joint	Joins two parts with a fixed joint.	Assembly with two parts or chosen faces/edges.
	revolute joint hinge revolute_joint	Joins two parts with a hinge (revolute joint).	Assembly with two parts or chosen faces/edges.
	slider joint slide parts slider_joint	Joins two parts with a slider joint.	Assembly with two parts or chosen faces/edges.
	distance joint hold apart distance_joint	Joins two parts keeping a distance.	Assembly with two parts or chosen faces/edges.
	angle joint angle between parts angle_joint	Joins two parts keeping an angle.	Assembly with two parts or chosen faces/edges.
	ball joint ball_joint	Joins two parts with a ball joint.	Assembly with two parts or chosen faces/edges.
	cylindrical joint cylinder joint cylindrical_joint	Joins two parts by their cylinders (rotation and sliding on the axis).	Assembly with two parts or chosen faces/edges.
	parallel joint keep parallel parallel_joint	Keeps two parts parallel.	Assembly with two parts or chosen faces/edges.
	perpendicular joint keep perpendicular perpendicular_joint	Keeps two parts perpendicular.	Assembly with two parts or chosen faces/edges.

Icon	Voice commands (English)	What does it do?	Requirements
	gears joint mesh gears gears_joint	Meshes two parts as gears.	Assembly with two parts or chosen faces/edges.
	belt joint belt_joint	Connects two parts with a belt.	Assembly with two parts or chosen faces/edges.
	screw joint screw_joint	Joins two parts with a screw joint.	Assembly with two parts or chosen faces/edges.
	rack and pinion joint rack_pinion_joint	Joins two parts as rack and pinion.	Assembly with two parts or chosen faces/edges.
	joint Group — enter by saying: joint	Joints that relate two parts by faces, edges or points.	Assembly with two parts or chosen faces/edges.
	angle joint angle angle	Sets the angle between two parts.	Active assembly.
	ball joint ball , sphere joint , sphere ball	Ball joint: the parts rotate freely about a point.	Active assembly.
	parallel joint parallel parallel	Keeps two parts parallel.	Active assembly.
	perpendicular joint perpendicular perpendicular	Keeps two parts perpendicular.	Active assembly.
	belt joint belt , chain joint , chain belt	Belt joint between two pulleys.	Active assembly.
	gear joint gear , gears gears	Gear joint between two gear wheels.	Active assembly.
	rack pinion rack and pinion , rack pinion joint , rack and pinion joint rackpinion	Rack-and-pinion joint.	Active assembly.
	screw joint screw , lead screw screw	Screw joint: rotation and advance are linked.	Active assembly.
	cylindrical joint cylindrical cylindrical	Cylindrical joint: rotation and sliding on the same axis.	Active assembly.
	distance joint distance distance	Sets the distance between two parts.	Active assembly.
	fixed joint fixed fixed	Fixed joint: the two parts are rigidly attached.	Active assembly.
	revolute joint revolute revolute	Revolute joint (hinge): rotation about one axis.	Active assembly.
	slider joint slider slider	Slider joint: one part slides along an axis.	Active assembly.

6. Workbench: Technical Drawings (TechDraw)

Enter with «workbench» → «technical drawing».

Icon	Voice commands (English)	What does it do?	Requirements
	views Group — enter by saying: views , drawing views	Projection, detail and section views.	Active TechDraw sheet.
	view view	Adds a view of the selected object to the sheet.	Active sheet and a selected 3D object.
	detail view detail , create detail view detailview	Creates an enlarged detail view of an area.	Active sheet with a view.
	broken view break view , create broken view brokenview	Creates a broken view (shortens a long part).	Active sheet with a view.
	clip group clipping group , create clip group clipgroup	Creates a clip group to show only one area.	Active TechDraw sheet.
	complex section section view , create section complexsection	Creates a complex section with several cutting planes.	Active sheet with a view.
	draft view draft , create draft view draft	Inserts a view of Draft objects.	Active TechDraw sheet.
	spreadsheet view spreadsheet , table view spreadsheet	Inserts a spreadsheet as a table on the drawing.	Active TechDraw sheet.
	view of object object view , view from object objectview	Creates a view of an object on the sheet choosing everything by voice: object (list or spelling), direction, scale and position.	Active TechDraw sheet.
	view direction change direction direction	Changes the projection direction of the view.	Active sheet with a view.
	view scale change scale scale	Changes the scale of the view.	Active sheet with a view.
	view position place view position	Moves the view to another position on the sheet.	Active sheet with a view.
	page projection sheet projection , projection type , third angle projection	Chooses the sheet's projection type (first or third angle).	Active TechDraw sheet.
	dimensions Group — enter by saying: dimensions , dimension , measurements	Length, radius, diameter, angle and other dimensions.	Active sheet with a view.
	dimension Group — enter by saying: dimension , size	General dimension.	Active TechDraw sheet.
	dimension measure dimension length , length dimension , length dimension	Adds a dimension based on the selection.	Active TechDraw sheet.

Icon	Voice commands (English)	What does it do?	Requirements
	measure Group — enter by saying: measure , length , distance	Length dimension.	Active TechDraw sheet.
	length length dimension , distance , measure , length	Dimensions the length of an edge.	Active TechDraw sheet.
	horizontal Group — enter by saying: horizontal , width , x distance	Horizontal dimension.	Active TechDraw sheet.
	horizontal horizontal dimension , width , x distance , horizontal	Dimensions the horizontal distance (width).	Active TechDraw sheet.
	extent Group — enter by saying: extent , span , total length	Overall extent dimensions.	Active TechDraw sheet.
	extent extent dimension , span , total length , extent	Dimensions the overall extent of the view.	Active TechDraw sheet.
	total dimensions overall dimensions , length and height , totals	Adds the overall dimensions (width and height) of the view.	Active TechDraw sheet.
	radius Group — enter by saying: radius , arc radius	Radius dimension.	Active TechDraw sheet.
	radius radius dimension , arc radius , radius	Dimensions the radius of an arc or circle.	Active TechDraw sheet.
	diameter Group — enter by saying: diameter , circle dimension	Diameter dimension.	Active TechDraw sheet.
	diameter circle dimension , diameter dimension , diameter	Dimensions the diameter of a circle.	Active TechDraw sheet.
	angle Group — enter by saying: angle , angular	Angle dimensions.	Active TechDraw sheet.
	angle angular , angle dimension , angle	Dimensions the angle between two lines.	Active TechDraw sheet.
	points three points , angle three points , 3 point angle , points	Dimensions an angle defined by three points.	Active TechDraw sheet.
	vertical vertical dimension , height , vertical	Dimensions the vertical distance (height).	Active TechDraw sheet.
	area area dimension , surface , area	Adds an area dimension to a face in the view.	Face selected in the view.
	tolerance hole shaft fit	Opens the panel to apply a standard ISO 286 fit.	Selected linear dimension.

Icon	Voice commands (English)	What does it do?	Requirements
	lines Group — enter by saying: lines , add lines	Auxiliary and center lines.	Active sheet with a view.
	two lines two line center , center line twolines	Center line between two lines.	Active TechDraw sheet.
	two points two point center , point center line twopoints	Center line between two points.	Active TechDraw sheet.
	cosmetic cosmetic line , construction line cosmetic	Cosmetic (auxiliary) line in the view.	Active TechDraw sheet.
	decorate line style , change line decorate	Changes a line's style.	Active TechDraw sheet.
	center face center , face center line center	Center line of a face.	Active TechDraw sheet.
	symmetry axis axis of symmetry , draw axis axis	Draws the axis of symmetry (center line) of a view chosen by voice.	Active TechDraw sheet.
	symbols Group — enter by saying: symbols , symbol	Welding symbols, rich text and surface finish.	Active sheet with a view.
	weld symbol welding symbol , weld weldsymbol	Inserts a welding symbol.	Active TechDraw sheet.
	rich text text annotation , annotation , formatted text richtext	Inserts rich text.	Active TechDraw sheet.
	surface finish finish symbol , surface finish symbol , surface roughness finish	Inserts the surface finish symbol.	Active TechDraw sheet.
	snaps Group — enter by saying: snaps , snap	Snaps to notable points.	Active sheet with a view.
	midpoint midpoints , middle point , center point midpoints	Snaps to midpoints.	Active TechDraw sheet.
	quadrant quadrants , quarter point , quadrant point quadrants	Snaps to quadrant points.	Active TechDraw sheet.
	topology Group — enter by saying: topology	Topological elements of the view.	Active sheet with a view.
	show all show everything , display all , reveal all showAll	Shows every topological element.	Active TechDraw sheet.
	page Group — enter by saying: page , sheet , drawing sheet	Sheets, templates and outputs (print, PDF, DXF, SVG).	Active document.

Icon	Voice commands (English)	What does it do?	Requirements
	default page new page , create page default	Creates a sheet with the default template.	Active TechDraw sheet.
	template page page template , new template page template	Creates a sheet with a chosen template.	Active TechDraw sheet.
	redraw page refresh page , update page redraw	Redraws and refreshes the sheet.	Active TechDraw sheet.
	print print page , print all print	Prints the active sheet.	Active TechDraw sheet.
	pdf export pdf , save pdf , print pdf pdf	Exports the active sheet to PDF.	Active TechDraw sheet.
	export dxf dxf , drawing exchange format , save dxf dxf	Exports the active sheet to DXF.	Active TechDraw sheet.
	export svg svg , save svg , export svg format svg	Exports the active sheet to SVG.	Active TechDraw sheet.
	annotations Group — enter by saying: annotations , notes	Annotations, balloons and axonometric dimensions.	Active TechDraw sheet.
	annotation note , text , comment annotation	Adds a text note to the sheet.	Active TechDraw sheet.
	axo length axonometric , axo dimension axo_length	Length dimension in an axonometric view.	Active sheet with a view.
	balloon text callout , bubble text balloon	Adds a balloon with a number or text.	Active sheet with a view.
	hatching Group — enter by saying: hatching , hatch	Hatching of faces.	Active sheet with a view.
	geometric hatch hatch , geometric pattern , pattern geometric_hatch	Applies a geometric hatch to a face.	Active TechDraw sheet.
	vertices Group — enter by saying: vertices , add vertices	Cosmetic vertices.	Active sheet with a view.
	cosmetic cosmetic vertex , reference point , add vertex cosmetic	Adds a cosmetic vertex (reference point).	Active TechDraw sheet.
	other views Group — enter by saying: other views , auxiliary views	Other views of the document.	Active TechDraw sheet.
	active view current view , view from screen , view from window active_view	Inserts the document's active view into the sheet.	Active TechDraw sheet.

Icon	Voice commands (English)	What does it do?	Requirements
	features Group — enter by saying: features	Template fields, images and symbols.	Active TechDraw sheet.
	fields template fields , fill fields fields	Fills in the template fields (title, author...).	Active TechDraw sheet.
	image insert image , add image image	Inserts an image into the sheet.	Active TechDraw sheet.
	symbol insert symbol , add symbol symbol	Inserts an SVG symbol into the sheet.	Active TechDraw sheet.

7. Workbench: PartDesign

Enter with «workbench» → «part design». The groups «base», «additive», «subtractive», «modify», «transform» and «manage» each have their own icon.

Icon	Voice commands (English)	What does it do?	Requirements
	base Group — enter by saying: base , basic , fundamental	Body, new sketch, clones and sub-shape binders.	Active document.
	body new body , create body body	Creates the Body that groups the features.	Active document.
	news sketch new sketch , create sketch news sketch	Creates a new sketch on a base plane or a flat face of the solid, chosen by voice.	Active document.
	clone clone feature , clone body , duplicate clone	Creates a dependent copy of a body or feature.	Selected object.
	subshapebinder subshape binder subshapebinder	References geometry of another body inside the active body.	Active Body.
	additive Group — enter by saying: additive , add , adding	Adds material: extrusions, revolutions, sweeps and primitives.	Active Body.
	pad pad feature , extrude , extrusion pad	Extrudes the selected sketch to create a solid (Pad).	Active Body and a closed sketch.
	revolution revolution feature revolution	Revolves the sketch about its vertical axis; lets you pick it by voice.	Active Body and a closed sketch.
	additive helix Coil feature additivehelix	Sweeps a profile along a helix (threads, springs).	Active Body and a closed sketch.
	additive loft loft feature additive loft	Creates a solid with a smooth transition between several profiles.	Active Body and two or more sketches.
	additive pipe pipe feature additivepipe	Sweeps a profile along a path.	Active Body and two or more sketches.
	additive box box , Additive solid box , box by size additivebox	Inserts an additive primitive box.	Active Body.
	additive cone cone , cone feature , cone by size additivecone	Inserts an additive cone or truncated cone.	Active Body.
	additive cylinder cylinder , cylinder feature , cylinder by size additivecylinder	Inserts an additive cylinder.	Active Body.
	additive ellipsoid ellipsoid feature additiveellipsoid	Inserts an additive ellipsoid.	Active Body.
	additive prism prism , prism feature , prism by size additiveprism	Inserts an additive regular prism.	Active Body.

Icon	Voice commands (English)	What does it do?	Requirements
	additive sphere sphere feature , sphere by radius additivesphere	Inserts an additive sphere.	Active Body.
	additive torus torus feature , torus by size addivetorus	Inserts an additive torus.	Active Body.
	additive wedge wedge feature additivewedge	Inserts an additive wedge.	Active Body.
	extrude sketch extend profile , thicken drawing pad_sketch	Extrudes the sketch and asks for the height by voice.	Active Body and a closed sketch.
	pad by length extrude by length , extrude by height , give height pad_by_length	Extrudes by a dictated length.	Active Body and a closed sketch.
	revolution by angle revolve by angle , spin profile revolve_by_angle	Revolves a sketch chosen by voice by the dictated angle.	Active Body and a closed sketch.
	blend shapes sweep surfaces , morph sections loft_profiles	Blends profiles with dictated sizes.	Active Body and two or more sketches.
	subtractive Group — enter by saying: subtractive , subtract , remove	Removes material: pockets, holes, grooves and negative primitives.	Active Body with a solid.
	pocket cut pocket	Removes material by extruding the sketch inward.	Solid in the Body and a closed sketch.
	groove slot groove	Removes material by revolving the sketch about an axis.	Solid in the Body and a closed sketch.
	hole drill hole	Creates threaded or plain holes from a sketch.	Solid in the Body and a closed sketch.
	subtractivebox cutbox , cut box by size subtractivebox	Subtracts a box from the solid.	Active Body with a solid.
	subtractivecone cutcone subtractivecone	Subtracts a cone from the solid.	Active Body with a solid.
	subtractivecylinder cutcylinder , cylinder , cut cylinder by size subtractivecylinder	Subtracts a cylinder from the solid.	Active Body with a solid.
	subtractiveellipsoid cutellipsoid subtractiveellipsoid	Subtracts an ellipsoid from the solid.	Active Body with a solid.
	subtractivehelix cuthelix , helix subtractivehelix	Removes material by sweeping a sketch along a helix.	Solid in the Body and a closed sketch.
	subtractiveloft cutloft subtractiveloft	Removes material by blending several profiles.	Solid in the Body and two or more sketches.
	subtractivepipe cutpipe subtractivepipe	Removes material by sweeping a sketch along a path.	Solid in the Body and two or more sketches.
	subtractiveprism cutprism subtractiveprism	Subtracts a polygonal prism from the solid.	Active Body with a solid.

Icon	Voice commands (English)	What does it do?	Requirements
	subtractivesphere cutsphere , cut sphere by radius subtractivesphere	Subtracts a sphere from the solid.	Active Body with a solid.
	subtractivetorus cuttorus subtractivetorus	Subtracts a torus from the solid.	Active Body with a solid.
	subtractivewedge cutwedge subtractivewedge	Subtracts a wedge from the solid.	Active Body with a solid.
	pocket by length hollow by length , hollow pocket_by_length	Pockets with a dictated depth.	Solid in the Body and a closed sketch.
	hole by size drill by size hole_by_size	Through hole: asks for the diameter.	Solid in the Body and a closed sketch.
	blind hole blind hole by size , blind drill blind_hole	Blind hole (does not go through): asks for diameter and depth and drills at each circle of the sketch.	Solid in the Body and a closed sketch.
	groove by angle groove_by_angle	Groove revolving the sketch by a dictated angle.	Solid in the Body and a closed sketch.
	boolean booleanoperation , booleanop boolean	Boolean operation between solids (union, intersection, difference).	Two bodies in the document.
	modify Group — enter by saying: modify , edit , editing	Fillets, chamfers, thickness, draft and text engraving.	Active Body with a solid.
	fillet round fillet	Rounds the selected edges.	One or more selected edges.
	chamfer bevel chamfer	Applies a chamfer to the selected edges.	One or more selected edges.
	draft taper draft	Tilts faces to ease mold release.	One or more selected faces.
	thickness shell thickness	Hollows the solid leaving walls of uniform thickness.	One or more selected faces.
	fillet by radius round edges fillet_by_radius	Rounds edges with the dictated radius.	One or more selected edges.
	chamfer by size bevel by size chamfer_by_size	Chamfer with the dictated size.	One or more selected edges.
	chamfer by size and angle chamfer with angle chamfer_by_size_and_angle	Chamfer with dictated size and angle.	One or more selected edges.
	thickness by value hollow solid thickness_by_value	Hollows the solid with the dictated thickness.	One or more selected faces.
	engrave engrave text , emboss , emboss text engrave	Engraves text raised or sunk: pick face or plane, raised or cut, spell the text and dictate heights.	Active Body with a solid.
	transform Group — enter by saying: transform , transformations , patterns	Linear and polar patterns, mirrors and scaling.	Active Body with a solid.

Icon	Voice commands (English)	What does it do?	Requirements
	linear pattern linear linearpattern	Repeats a feature along a straight line.	Selected feature.
	mirrored mirror mirrored	Mirrors the selected features.	Selected feature.
	polar pattern polar , circular pattern polarpattern	Repeats a feature in a circle.	Selected feature.
	multi transform multi multitransform	Combines several successive transformations.	Selected feature.
	scaled scale , resize scaled	Creates progressively scaled copies.	Selected feature.
	linear pattern by length repeat in line linear_pattern	Linear pattern with the dictated count and size.	Selected feature.
	linear pattern by spacing repeat every linear_pattern_by_spacing	Linear pattern with dictated spacing.	Selected feature.
	polar pattern by angle repeat in circle polar_pattern	Circular pattern with dictated count and angle.	Selected feature.
	scale by factor scale solid scaled_by_factor	Scaled copies with the dictated factor.	Selected feature.
	manage Group — enter by saying: manage , management , administration	Move features, the body tip and wizards.	Active Body.
	movefeature move feature movefeature	Moves an object from one body to another.	Two bodies in the document.
	movefeatureintree move feature in tree movefeatureintree	Reorders features within a body's tree.	Active Body.
	movetip move tip movetip	Sets a feature as the body's Tip.	Active Body.
	preferences settings preferences	Opens the PartDesign preferences.	None.
	wizardshaft wizard shaft wizardshaft	Creates a stepped shaft by voice: you dictate the number of sections and each one's length and diameter (axis along Z).	None.

8. Workbench: Sketcher

Enter with «workbench» → «sketcher».

Icon	Voice commands (English)	What does it do?	Requirements
	root origin , center , center point root	Selects the sketch root point (origin).	Open sketch (in edit mode).
	import image reference image , background image , bring image image	Places an image behind the sketch to trace it.	None.
	new new sketch , create sketch new	Creates a new sketch.	Active document.
	close sketch leave sketch , exit sketch , finish sketch leave	Closes the sketch being edited, keeping the drawing.	Open sketch (in edit mode).
	edit edit sketch , modify sketch edit	Edits the selected sketch.	Selected sketch.
	attach map sketch , attach sketch attach	Attaches or maps the selected sketch.	Selected sketch.
	grid toggle grid , show grid grid	Shows or hides the grid.	Open sketch (in edit mode).
	cancelediting cancel editing , stop editing cancelediting	Leaves editing, canceling the changes.	Open sketch (in edit mode).
	carboncopy carbon copy carboncopy	Copies the geometry of another sketch into the active one.	Open sketch (in edit mode).
	copyelements copy elements , copy geometry copyelements	Copies the selected elements to the clipboard.	Geometry selected in the sketch.
	cutelements cut elements , cut geometry cutelements	Cuts the selected elements.	Geometry selected in the sketch.
	pasteelements paste elements , paste geometry pasteelements	Pastes the clipboard elements into the sketch.	Open sketch (in edit mode).
	mirror symmetry , mirror elements mirror	Mirrors the selected elements about a line or axis.	Geometry selected in the sketch.
	mirrorsketch mirror sketch mirrorsketch	Mirrors the whole sketch about the main axes.	Open sketch (in edit mode).
	offset create offset offset	Creates geometry parallel at a distance from the selection.	Geometry selected in the sketch.
	movearray translate , move elements movearray	Moves or array-copies the selected geometry.	Geometry selected in the sketch.
	rotatepolar rotate , rotate elements rotatepolar	Rotates or polar-copies the selected geometry.	Geometry selected in the sketch.

Icon	Voice commands (English)	What does it do?	Requirements
	scale scale elements scale	Changes the scale of the selected geometry.	Geometry selected in the sketch.
	trimedge trim , trim edge trimedge	Trims an edge using other geometry as the limit.	Open sketch (in edit mode).
	splitedge split , split edge splitedge	Splits an edge in two at the chosen point.	Open sketch (in edit mode).
	extendedge extend , extend edge extendedge	Extends an edge up to other geometry.	Open sketch (in edit mode).
	fillet create fillet fillet	Creates a fillet between two adjacent edges.	Open sketch (in edit mode).
	chamfer create chamfer chamfer	Creates a chamfer between two adjacent edges.	Open sketch (in edit mode).
	geometry Group — enter by saying: geometry , geometric	Sketch geometric entities and leaving the sketch.	Open sketch (in edit mode).
	arc Group — enter by saying: arc	Arcs.	Open sketch (in edit mode).
	center center arc center	Arc by center, radius and start and end angles.	Open sketch (in edit mode).
	three point three points , 3 point , 3 points 3point	Arc by two endpoints and a middle point.	Open sketch (in edit mode).
	arc by center create arc by center , arc by angles , arc by coordinates create_by_center	Arc by dictated center and angles.	Open sketch (in edit mode).
	arc by three points create arc by three points , three point arc by coordinates create_by_3points	Arc by three dictated points.	Open sketch (in edit mode).
	arc slot Group — enter by saying: arc slot	Curved slots.	Open sketch (in edit mode).
	arc ends rounded slot , curved slot , arc slot arc_ends	Curved slot with rounded ends.	Open sketch (in edit mode).
	flat ends flat slot , square ends , flat arc slot flat_ends	Curved slot with flat ends.	Open sketch (in edit mode).
	bspline Group — enter by saying: bspline , spline	B-Spline curves.	Open sketch (in edit mode).
	create create bspline , create b-spline , bspline curve create	B-Spline by control points.	Open sketch (in edit mode).
	interpolation interpolated , interpolated bspline interpolation	B-Spline passing exactly through the points.	Open sketch (in edit mode).
	periodic periodic bspline , closed bspline periodic	Periodic (closed) B-Spline by control points.	Open sketch (in edit mode).

Icon	Voice commands (English)	What does it do?	Requirements
	periodicinterp periodicinterp	Closed B-Spline interpolating the points.	Open sketch (in edit mode).
	tools Group — enter by saying: tools , bspline tools	Tools to edit B-Splines.	B-Spline curve selected in the sketch.
	tonurbs convert to nurbs tonurbs	Converts the curve to NURBS.	Open sketch (in edit mode).
	decrease degree down decrease	Lowers the B-Spline degree by one.	Open sketch (in edit mode).
	increase degree up increase	Raises the B-Spline degree by one.	Open sketch (in edit mode).
	knot insert knot knot	Inserts a knot into the B-Spline.	Open sketch (in edit mode).
	join join curve join	Joins two continuous B-Spline curves.	Open sketch (in edit mode).
	circle Group — enter by saying: circle	Circles.	Open sketch (in edit mode).
	create circle , center circle , circle by center create	Circle by center and radius (voice window).	Open sketch (in edit mode).
	three point three points , 3 point , 3 points 3point	Circle through three points.	Open sketch (in edit mode).
	circle by parameters circle by center and radius create_by_center_radius	Circle with dictated center and radius.	Open sketch (in edit mode).
	parametric three point circle three points parametric create_by_3points	Circle by three dictated points.	Open sketch (in edit mode).
	ellipse Group — enter by saying: ellipse	Ellipses and conics.	Open sketch (in edit mode).
	center ellipse center , ellipse by center , create ellipse by center center	Ellipse by center and major and minor radii.	Open sketch (in edit mode).
	3points 3 points , three points , three point 3points	Ellipse by three points.	Open sketch (in edit mode).
	elliptic ellipse arc , elliptic arc by coordinates elliptic	Elliptical arc with dictated angles.	Open sketch (in edit mode).
	hyperbolic hyperbola arc , hyperbolic arc by coordinates hyperbolic	Hyperbola arc.	Open sketch (in edit mode).
	parabolic parabola arc , parabolic arc by coordinates parabolic	Parabola arc.	Open sketch (in edit mode).
	heptagon Group — enter by saying: heptagon	Heptagons.	Open sketch (in edit mode).

Icon	Voice commands (English)	What does it do?	Requirements
	create create heptagon , draw heptagon , heptagon create	Regular heptagon by center and radius.	Open sketch (in edit mode).
	hexagon Group — enter by saying: hexagon	Hexagons.	Open sketch (in edit mode).
	create create hexagon , draw hexagon , hexagon create	Regular hexagon by center and radius.	Open sketch (in edit mode).
	line Group — enter by saying: line , straight line	Lines.	Open sketch (in edit mode).
	create create line , draw line , line create	Straight line between two points.	Open sketch (in edit mode).
	line by points create line by points , draw line by points , line by coordinates create_by_points	Line with the four dictated coordinates (x1, y1, x2, y2).	Open sketch (in edit mode).
	polygon Group — enter by saying: polygon	Regular polygons.	Open sketch (in edit mode).
	pentagon regular pentagon , create pentagon , five sides pentagon	Regular pentagon.	Open sketch (in edit mode).
	octagon regular octagon , create octagon , eight sides octagon	Regular octagon.	Open sketch (in edit mode).
	regular polygon , regular polygon , polygon by sides regular	Regular polygon with the dictated number of sides.	Open sketch (in edit mode).
	polyline Group — enter by saying: polyline , multi line	Polylines.	Open sketch (in edit mode).
	create create polyline , draw polyline , polyline create	Line made of several segments.	Open sketch (in edit mode).
	rectangle Group — enter by saying: rectangle , box	Rectangles.	Open sketch (in edit mode).
	create create rectangle , draw rectangle , rectangle create	Rectangle by two opposite corners.	Open sketch (in edit mode).
	center center rectangle , centered rectangle , create centered rectangle center	Rectangle by center and size.	Open sketch (in edit mode).
	triangle Group — enter by saying: triangle	Triangles.	Open sketch (in edit mode).
	create create triangle , draw triangle , triangle create	Triangle by its three vertices.	Open sketch (in edit mode).
	new new sketch , create sketch new	Creates a new sketch.	Active document.
	close sketch leave sketch , exit sketch , finish sketch leave	Closes the sketch being edited, keeping the drawing.	Open sketch (in edit mode).

Icon	Voice commands (English)	What does it do?	Requirements
	edit edit sketch edit	Edits a sketch by dictating its name.	Active document.
	attach map sketch attach	Attaches a sketch to a dictated face or support.	Active document.
	grid grid	Shows or hides the sketch grid.	Open sketch (in edit mode).
	arcslot Group — enter by saying: arcslot , arcs , arc	Arc slots.	Open sketch (in edit mode).
	arc slot curved ends , rounded slot , slot with arcs arcends	Curved slot with rounded edges.	Open sketch (in edit mode).
	constraints Group — enter by saying: constraints , constraint , restrictions	Dimensional (size) and geometric constraints.	Open sketch (in edit mode).
	dimension add dimension , length dimension	Distance dimension on a line.	Open sketch (in edit mode).
	horizontal horizontal dimension , horizontal distance horizontal	Horizontal dimension between a line's endpoints.	Open sketch (in edit mode).
	vertical vertical dimension , vertical distance vertical	Vertical dimension between a line's endpoints.	Open sketch (in edit mode).
	angle angle dimension angle	Angular dimension between two lines.	Open sketch (in edit mode).
	radius radius dimension radius	Radius dimension of an arc or circle.	Open sketch (in edit mode).
	diameter diameter dimension , radiam , auto dimension diameter	Diameter dimension of a circle.	Open sketch (in edit mode).
	distance distance dimension distance	Distance dimension between a line's endpoints.	Open sketch (in edit mode).
	geometric Group — enter by saying: geometric , geometric constraints , geometry constraints	Geometric relations between entities.	Open sketch (in edit mode).
	coincident make coincident , join points coincident	Makes two points coincide.	Two sketch points.
	coincident unified unified coincident coincidentunified	Automatic coincidence over several points.	Two sketch points.
	lock fix position lock	Locks a point at absolute coordinates.	A sketch point.
	point on object fix to line pointonobject	Forces a point to lie on a line or curve.	A point and a line or curve.
	horizontal make horizontal horizontal	Forces a line to be horizontal.	One or more sketch lines.

Icon	Voice commands (English)	What does it do?	Requirements
	vertical make vertical vertical	Forces a line to be vertical.	One or more sketch lines.
	horizontal vertical auto orientation horver	Applies horizontal or vertical depending on the slope.	One or more sketch lines.
	parallel make parallel parallel	Makes two or more lines parallel.	Two or more sketch lines.
	perpendicular make perpendicular perpendicular	Makes two lines perpendicular (90°).	Two or more sketch lines.
	tangent make tangent tangent	Makes two curves, or a curve and a line, tangent.	Two curves or a curve and a line.
	equal make equal , equal length equal	Equalizes the length or radius of two entities.	Two curves or a curve and a line.
	symmetric make symmetric , symmetry , apicate symmetry symmetric	Makes two points symmetric about a line or point.	Two sketch points.
	block block geometry block	Locks the geometry in its current position.	Geometry selected in the sketch.
	toggle driving reference mode , driving mode toggledriving	Toggles between driving and reference constraint.	Selected constraint.
	toggle active activate constraint , deactivate constraint toggleactive	Enables or disables the selected constraints.	Selected constraint.
	external Group — enter by saying: external	External geometry from other parts.	Open sketch and visible external geometry.
	projection project , project edge projection	Projects an external edge onto the sketch.	Open sketch (in edit mode).
	intersection intersect , external geometry , external link intersection	Intersects external geometry with the sketch.	Open sketch (in edit mode).
	oblong Group — enter by saying: oblong , create oblong	Oblongs.	Open sketch (in edit mode).
	create create slot , draw slot , slot create	Oblong (capsule) by the two centers and radius.	Open sketch (in edit mode).
	point Group — enter by saying: point , create point	Points.	Open sketch (in edit mode).
	create create point , draw point , point create	Creates a sketch point.	Open sketch (in edit mode).
	create point by coordinates point by coordinates , vertex by coordinates create_by_coords	Point with the dictated X, Y coordinates.	Open sketch (in edit mode).
	select Group — enter by saying: select , selection	Selects the sketch axes and origin.	Open sketch (in edit mode).

Icon	Voice commands (English)	What does it do?	Requirements
	horizontal horizontal axis , select horizontal horizontal	Selects the horizontal X axis.	Open sketch (in edit mode).
	vertical vertical axis , select vertical vertical	Selects the vertical Y axis.	Open sketch (in edit mode).
	origin center point , select origin , zero point origin	Selects the origin (0, 0).	Open sketch (in edit mode).
	slot Group — enter by saying: slot , create slot	Slots.	Open sketch (in edit mode).
	create slot , oblong , create slot create	Slot made of two arcs and two parallel lines.	Open sketch (in edit mode).
	square Group — enter by saying: square , create square , draw square	Squares.	Open sketch (in edit mode).
	create create square , draw square , square create	Axis-aligned square by center and radius.	Open sketch (in edit mode).
	text Group — enter by saying: text , write text , create text	Text in the sketch.	Open sketch (in edit mode).
	create create text , shape string , add text shapestring	Creates a text shape in the sketch.	Open sketch (in edit mode).
	tools Group — enter by saying: tools , tool	Close, clear, merge and reorient sketches.	Open sketch (in edit mode).
	close sketch leave sketch , exit sketch , finish sketch leave	Closes the sketch being edited, keeping the drawing.	Open sketch (in edit mode).
	deleteconstraints delete constraints , clear constraints , remove constraints deleteconstraints	Removes every constraint of the sketch.	Open sketch (in edit mode).
	deletegeometry delete geometry , clear geometry , clear sketch deletegeometry	Deletes all geometry and constraints of the sketch.	Open sketch (in edit mode).
	merge merge sketches , combine sketches merge	Merges the geometry of several sketches into one.	Two or more selected sketches.
	reorient reorient sketch , change sketch plane reorient	Reorients the selected sketch to another face.	Selected sketch.
	removeaxes remove axes , clear axes alignment removeaxes	Removes the automatic axis alignment.	Selected sketch.
	validate Group — enter by saying: validate , validate sketch	Sketch validation.	Open sketch (in edit mode).
	validate validate sketch , check sketch , verify validate	Checks the sketch, finds missing coincidences and fixes tolerances.	Open sketch (in edit mode).
	view Group — enter by saying: view , view sketch , view selection	Sketch views.	Open sketch (in edit mode).

Icon	Voice commands (English)	What does it do?	Requirements
	sketch sketch view , view sketch , look at sketch sketch	Orients the camera perpendicular to the sketch.	Open sketch (in edit mode).
	section section view , view section , clip section section	Toggles the sketch's section view.	Open sketch (in edit mode).

9. Workbench: Draft

Enter with «workbench» → «draft».

Icon	Voice commands (English)	What does it do?	Requirements
	import image reference image , background image , bring image image	Places a reference image on the XY, XZ or YZ plane.	Active working plane.
	annotation Group — enter by saying: annotation , note , text	Texts and labels.	Active working plane.
	text simple text , add text , write text text	Inserts 2D text on the working plane.	Active working plane.
	shape string shapestring , 3d text , physical text shapestring	Creates a flat shape from text and a font.	Active document and a font file.
	label add label , tag , callout label	Creates a label with a leader line.	Active working plane.
	annotation styles Group — enter by saying: annotation styles , style editor	Annotation styles.	Active working plane.
	editor style editor , annotation editor , annotation style editor editor	Opens the text, dimension and label style editor.	Active document.
	arc Group — enter by saying: arc	Circular arcs.	Active working plane.
	center center arc , arc by center , standard arc center	Arc by center, radius and angles.	Active working plane.
	points 3 points , three points , arc by 3 points points	Arc through three points.	Active working plane.
	curve Group — enter by saying: curve , spline , b-spline	Smooth curves.	Active working plane.
	bezier bezier curve , n degree bezier bezier	Degree-N Bézier curve.	Active working plane.
	bspline b spline , spline , smooth curve bspline	B-Spline curve following the given points.	Active working plane.
	cubic cubic bezier , cubic curve , third degree bezier cubic	Cubic Bézier curve.	Active working plane.
	circle Group — enter by saying: circle	Circles.	Active working plane.
	center circle , create circle , standard circle center	Circle by center and radius.	Active working plane.
	array Group — enter by saying: array , pattern , circular array	Arrays: repeat an object in a pattern.	Selected object.

Icon	Voice commands (English)	What does it do?	Requirements
	circular circular array , concentric array circular	Circular array: copies in concentric rings.	Active working plane.
	ortho ortho array , orthogonal , orthogonal array ortho	Orthogonal array: rows and columns.	Active working plane.
	polar polar array polar	Polar array: copies around a center.	Active working plane.
	path path array , along path , array along curve path	Array along a path.	Object and path selected.
	path link path link array , link path array , link along path pathlink	Path array made of links (lighter).	Object and path selected.
	point point array , array by points point	Array placing copies at given points.	Object and points selected.
	point link point link array , link point array , link by points pointlink	Point array made of links.	Object and points selected.
	modify Group — enter by saying: modify , edit , change	Operations on existing shapes.	Selected object.
	clone copy clone	Makes a linked copy: if the original changes, so does the clone.	Active working plane.
	downgrade simplify downgrade	Simplifies a shape (for example, splits a face into lines).	Active working plane.
	sketch convert to sketch sketch	Converts a Draft shape into a sketch, or back.	Active working plane.
	edit modify edit	Enters edit mode to move the shape's points.	Active working plane.
	fillet round fillet	Creates a fillet between two lines or segments.	Active working plane.
	join combine join	Joins two or more lines into a single object.	Active working plane.
	cut subtract , cut out cut	Cuts one object with another: removes from the first the part the second occupies.	Two or more selected objects.
	move translate move	Moves or copies objects to a new position.	Active working plane.
	offset parallel copy offset	Creates a parallel copy at a given distance.	Active working plane.
	rotate turn rotate	Rotates or copies objects around a center.	Active working plane.

Icon	Voice commands (English)	What does it do?	Requirements
	mirror reflect mirror	Creates the mirrored copy about a line.	Active working plane.
	measurement Group — enter by saying: measurement	Draft dimensions.	Active working plane.
	dimension linear dimension , measure linear	Creates a linear dimension with its measurement.	Active working plane.
	flip dimension flip , reverse dimension flip	Flips the dimension text by rotating it 180°.	Selected Draft dimension.
	ellipse Group — enter by saying: ellipse , oval	Ellipses.	Active working plane.
	ellipse center ellipse , oval center	Ellipse by major and minor radii.	Active working plane.
	facebinder Group — enter by saying: facebinder , binder , face binder	Surfaces from faces.	Active working plane.
	facebinder create facebinder , create create	Creates a surface from the selected faces.	One or more selected faces.
	drafting Group — enter by saying: drafting , draw	Lines and polylines.	Active working plane.
	wire polyline , continuous line , multi line wire	Draws a multi-segment polyline.	Active working plane.
	line line by points , draw line , segment line	Draws a straight line, also «line by points».	Active working plane.
	creation Group — enter by saying: creation , create , create object	Basic shapes: point, polygon, rectangle, triangle and hatch.	Active working plane.
	hatch fill pattern , hatch pattern , pattern hatch	Fills a shape with a pattern.	Selected object.
	point create point , dot point	Marks a point at the dictated coordinates.	Active working plane.
	polygon regular polygon , create polygon polygon	Draws a regular polygon.	Active working plane.
	rectangle create rectangle , draw rectangle , box rectangle	Draws a rectangle by two corners.	Active working plane.
	rectangle by center centered rectangle , center rectangle , create centered rectangle rectangle_center	Draws a rectangle by its center and size.	Active working plane.
	triangle create triangle , draw triangle , triangle by vertices triangle	Draws a triangle by its three vertices.	Active working plane.
	modification Group — enter by saying: modification , modifications	More operations on existing shapes.	Selected object.

Icon	Voice commands (English)	What does it do?	Requirements
	scale resize scale	Enlarges or shrinks proportionally.	Active working plane.
	shape 2d view 2d view shape_2d_view	Generates a flat 2D projection of a 3D object.	Active working plane.
	slope incline slope	Changes the slope (Z inclination) of a wire.	Active working plane.
	split divide split	Splits a line or wire at a point.	Active working plane.
	stretch extend stretch	Stretches geometry by moving a set of vertices.	Active working plane.
	subelement highlight highlight subelement_highlight	Highlights faces, edges or vertices of the selection.	Active working plane.
	trimex trim trimex	Trims or extends an entity up to another.	Active working plane.
	upgrade improve upgrade	Promotes simple elements to faces or solids.	Active working plane.
	wire to bspline bspline wire_to_bspline	Converts a polyline into a smooth B-Spline curve.	Active working plane.
	pointplacement Group — enter by saying: pointplacement , point placement , place point	Placing reference points.	Active document.
	add point addpoint addpoint	Places a point at the selected vertex.	Selected vertex or object with vertices.
	point at vertex pointatvertex pointatvertex	Inserts a point at the given vertex index.	Selected vertex or object with vertices.
	midpoint mid point midpoint	Places a point at the middle of the selected edge.	One or more selected edges.
	point at coordinates point at coords , pointatcoords pointatcoords	Creates a point at exact X, Y, Z coordinates.	Active working plane.
	pointconnect Group — enter by saying: pointconnect , point connect , connect points	Connect points to form lines and polygons.	Object and points selected.
	connect connect points , join points connect	Connects the selected points.	Active working plane.
	create line make line createline	Creates a line between the points.	Active working plane.
	close wire close cable closewire	Closes the wire joining the last point to the first.	Active working plane.
	create polygon make polygon createpolygon	Creates a polygon from the points.	Active working plane.

10. Workbench: Part

Solids, primitives and CSG operations. Enter with «workbench» → «part».

Icon	Voice commands (English)	What does it do?	Requirements
	box block , cube , add cube box	Creates a parametric box (10 × 10 × 10 mm by default).	Active document.
	circle round , add round , add circle circle	Creates a circle or arc as an edge (not a solid).	Active document.
	cone primitive cone , add cone , add primitive cone cone	Creates a primitive cone or truncated cone.	Active document.
	cube box , block , add cube cube	Creates a parametric solid cube (10 × 10 × 10 mm).	Active document.
	cylinder primitive cylinder , tube , add cylinder cylinder	Creates a parametric cylinder (radius 2 mm, height 10 mm).	Active document.
	ellipse oval , add oval , add ellipse ellipse	Creates an ellipse or elliptical arc as an edge.	Active document.
	line straight , add line , add straight line	Creates a straight segment between two 3D points.	Active document.
	sketch new sketch , create sketch , make sketch new sketch	Creates a new sketch on a plane or a face.	Active document.
	chamfer create chamfer , make chamfer , bevel chamfan	Applies a chamfer to the selected edges.	One or more selected edges.
	color per face paint face , color face , set face color paint face	Paints each face of the object a different color.	Selected Part object.
	cross sections create cross sections , make cross sections , cross section secciones transversales	Generates cross sections of the object.	Selected Part object.
	extrude create extrusion , make extrusion , extrude object extruir	Extrudes a profile and creates a solid.	Selected sketch or outline.
	fillet create fillet , make fillet , round redondear bordes	Rounds the selected edges.	One or more selected edges.
	loft create loft , make loft , join profiles hacer loft	Creates a solid or surface between several profiles (loft).	Two or more selected profiles.
	make face makeface , create face , createface makeface	Turns a closed outline into a face.	Selected sketch or outline.
	upgrade upgrade	Automatically converts a closed wire into a face.	Selected sketch or outline.
	mirror reflect , create mirror espejo	Creates the mirrored copy of the object about a plane.	Selected Part object.

Icon	Voice commands (English)	What does it do?	Requirements
	offset create offset , thicken , shrink desfase	Thickens or offsets the object 1.0 mm in 3D.	Selected Part object.
	offset 2d 2d offset , outline , border contorno	Offsets a flat wire or face by 1.0 mm.	Selected 2D wire or face.
	projection project , project on surface proyectar dibujo	Projects shapes onto a support surface.	Support surface and at least one shape.
	revolve create revolve , revolution revolucion	Generates a 360° solid of revolution about the Z axis.	Selected sketch or outline.
	ruled surface create ruled surface , join curves superficie reglada	Creates a ruled surface between two edges or curves.	Exactly two edges or curves.
	scale resize , enlarge , reduce escalar	Scales the object by a uniform factor of 2×.	Selected Part object.
	section create section , make section , section curve intersección	Computes the intersection curve between two objects.	Exactly two Part objects.
	sweep create sweep , make sweep , sweep profile barrer perfil	Generates a solid by sweeping a profile along a path.	Profile and guide path selected.

Guided examples

Learning by doing

DAV ships with examples that are built step by step, by voice. Each frame tells you what to say; when you have said it all, in order, the action runs and the next frame appears.

How to open an example

1. Say **explorer**, then **examples** and **examples** again: the picker opens.
2. With **next** (or **back**) go through the list to the example and say **send**.
3. A window, which does not block FreeCAD, shows one frame at a time with what you would say to do it: first the path through the menus and then the values. Each value is confirmed with **send**.
4. **skip** runs the frame without saying its words; **cancel** closes the example; at the end, **send** closes it.
5. Numbers from 0 to 99 are said naturally («thirty two»); from 100 on, digit by digit. Decimals use «point».

Available examples

Example	What is done
Sketcher: a circle with a constraint and a dimension	A circle with a radius constraint and a 2D dimension.
Draft: 2D drawing with dimensions	Rectangle, circle and polygon with dictated sizes.
Draft: a house from 2D shapes	A house from 2D shapes with dictated sizes; you learn to trim one shape with another.
Draft: simple title block with text	A simple title block with text in Draft.
TechDraw: a circle with its title block	A circle on a TechDraw sheet with its title block.
PartDesign: a screw, step by step	A screw: shank, tip, head, chamfer and thread; 3D dimension and «isometric» view.
Die	A 20 mm die: the 1 face with a cylinder and the other five with a sketch and a pocket each.
M6 flat washer	An M6 washer: sketch with the hole (Ø 6.4) and the edge (Ø 12), extruded 1.6 mm and placed on a TechDraw sheet.
Bolt-nut	An M6 bolt and its nut in PartDesign, joined in an assembly with a cylindrical joint.
Five-part scissors	Two blades, two handles and a bolt joined in an assembly with hinges and fixed joints.

House: 2D shapes (Draft)

A house is drawn with 2D shapes only, with dictated sizes, and you learn to trim one shape with another.

Body	rectangle by corners, from (2, 0) to (42, 25)
Roof	triangle by vertices (0, 25), (44, 25) and (22, 50)
Door and windows	rectangle by corners from (17, 0) to (27, 14); two «rectangle by center» of 8 × 8 at (10, 14) and (34, 14)
Chimney	rectangle from (31, 30) to (37, 48) and a triangle as its cap; trimmed with modify → cut
Details	round window (circle at (22, 34), radius 5), door knob (radius 1) and the window crosses with lines by points

Path: workbench → draft → create → rectangle / triangle / circle, and up → drawing → line for the lines.

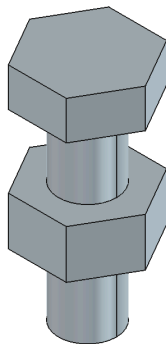


Result of the house example

Bolt and nut: an assembly (PartDesign and Assembly)

An M6 bolt (simplified from DIN 931) and its nut, made in PartDesign and joined in an assembly.

Shank	cylinder with radius 3 and height 20, centered at (0, 0, 10)
Head	6-sided prism, radius 6 and height 4, centered at (0, 0, 22), added to the same body
Nut	another body: 6-sided prism, radius 6 and height 5, at (15, 0, 4), with a subtractive cylinder of radius 3
Assembly	create assembly, insert link twice, ground part (the bolt) and a cylindrical joint through the shank and hole cylinders; solve and the «isometric» view



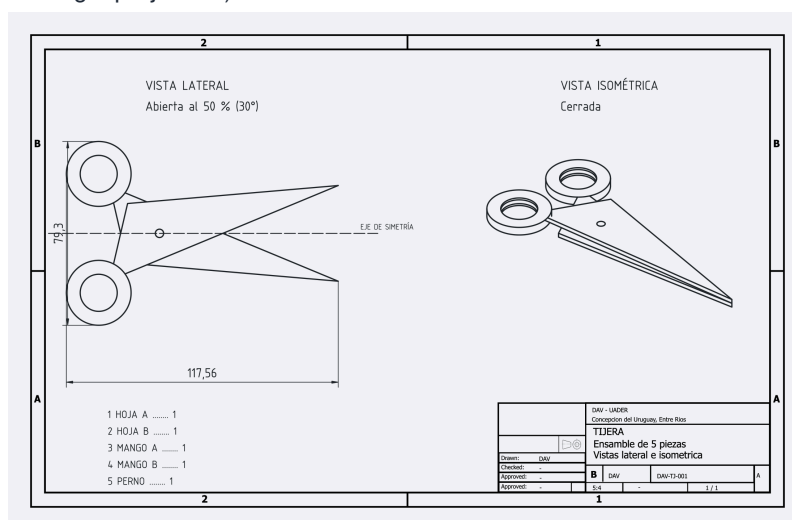
Bolt and nut, isometric view

Five-part scissors: assembly and drawing (PartDesign and Assembly)

Two blades, two handles and a bolt: five parts in PartDesign, joined in an assembly with hinges and fixed joints.

Blades	triangle by vertices (80, 0), (-10, -16), (-26, 14) extruded 2 mm, plus a tab and two holes; blade B is its mirror in Y
Handles	rings of radius 14 and height 4, with a hole of radius 8, at (-32, ±18)
Bolt	stepped (radius 2 and 1.8): this way the assembly stacks blade B 2 mm above A without position joints
Assembly	5 × insert link, ground part (the bolt), two hinge (blades) and two fixed joint (blade-handle); solve and «isometric»

The DAV example goes as far as the closed scissors. The guide <Dav/docs/guia-tijeras-voz.md> continues with the scissors open at 50 % (a distance joint of 23.3 mm between the handles) and the technical drawing on an ANSI B sheet (ASME Y14.1, third-angle projection).



Scissors drawing: side view open at 50 % and closed isometric view

File locations in the repository

Command dictionaries	<i>Dav/dic/</i> : each folder has its <i>TraduceToEs.py</i> (Spanish), <i>TraduceToEn.py</i> (English) and <i>TraduceToPt.py</i> (Portuguese).
SVG icons	Next to each command, named after its key (for example <i>save.svg</i>); the panel's own icons are in <i>Dav/scr/ComponentesDAV/InterfazDAV/Icons/</i> .
Selection engine	<i>Dav/scr/selection/object_selection.py</i> .
Global navigation	<i>Dav/dic/NavCommands/</i> (up, context, send, cancel) and <i>Dav/dic/Correction/</i> (undo, delete).
Guides and manuals	<i>Dav/docs/</i> . This manual is generated with <i>Dav/docs/manual/build_manual.py</i> from the real dictionaries.

Acknowledgements

We would like to extend our special thanks to the [Autonomous University of Entre Ríos \(UADER\)](#) for providing us with the space and resources necessary to carry out this project; to professors [Jesús Valenzuela](#) and [Bernabe Arias](#), for their support, guidance, and willingness throughout the process; and to communicators [Naitria Peralta Montoya](#) and [Bruno Contigiani](#), for their collaboration and contributions to the development and dissemination of the project.

To everyone who, in one way or another, contributed to making this project possible, thank you very much.